

Protection without Power: The Macroeconomic and Spatial Determinants of Market Power and Misallocation in Indian Manufacturing, 2012–2022*

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Abstract

Does protection buy market power, and does growth heal or aggravate misallocation? We assemble an eleven-year determinants panel for Indian organised manufacturing — plant-level markups, markdowns, monopsony incidence, and revenue-productivity dispersion aggregated to 23 industries, 69 industry groups, and 36 states — and merge it with a newly concorded applied-tariff panel (WITS HS-6 lines mapped to NIC-2008 through the official UN correspondence chain, 2011–2023), BIS real exchange rates interacted with predetermined import exposure, state real GSDP, and local labour-market concentration. Identification combines two-way fixed-effects shift-share designs with a year-demeaned panel vector autoregression. Four findings. First, *protection does not buy power*: every tariff–markup coefficient across both industry panels is negative — significant under import-weighted tariffs, outside COVID years, and with national controls — and the panel VAR shows tariff shocks lowering markups for one to four years; this is the incomplete-pass-through result of De Loecker et al. (2016) running in reverse through the 2018+ protectionist turn. Second, *the exchange rate disciplines*: real appreciation compresses markups in import-exposed industries at both industry levels — textbook import discipline, now operating through prices rather than tariff lines. Third, *misallocation is domestic*: it is orthogonal to trade policy in all specifications, yet within

*Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee. This paper is the third of a series based on a newly constructed plant panel from the Annual Survey of Industries; the companion papers measure markups and markdowns (working paper 1) and misallocation under variable markups (working paper 2). ASI unit records are used under licence from the Ministry of Statistics and Programme Implementation and cannot be redistributed; all aggregate panels, the tariff concordance, and code are available from the authors. All errors are ours.

states it *rises* robustly with real GSDP — the signature of transition dynamics, not border distortions. Fourth, *tariff-setting is endogenous*: markups and misallocation Granger-cause tariffs, so the decade’s protection episode tracked industry conditions — a feedback any evaluation of Indian trade policy must confront. Space, not macro policy, is where labour-market power lives: agglomerated states are more competitive on every margin, and the rural–urban monopsony gap is the sharpest spatial fact in the data.

JEL codes: F13, F14, L11, J42, O14, O25, R12.

Keywords: tariffs; import discipline; real exchange rate; markups; monopsony; misallocation; panel VAR; shift-share; Indian manufacturing.

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1 Introduction

Two prior papers in this series established, from the plant records of the Annual Survey of Industries, that product-market power in Indian organised manufacturing is modest and stable (median markups near 1.22, drifting down), that labour-market power is pervasive (two plants in five pay labour below its marginal revenue product), and that within-industry misallocation is large but declining, carried by capital wedges rather than markups. Those papers treated the macroeconomic environment as background. This paper brings the environment to the foreground and asks the natural next question: *what moves these objects?* Which policies and conditions — tariffs, the exchange rate, state growth, labour-market structure, geography — determine where market power and misallocation rise and fall?

The question is not merely descriptive; each candidate determinant carries a theoretical prediction that the data can confront. Consider tariffs. In the Melitz–Ottaviano class of models, protection shields incumbents from import competition, and markups *rise*. But protection also raises the cost of imported inputs, and when pass-through of cost into price is incomplete — the empirical rule in producer-level data — measured markups can *fall* with protection: this is precisely the mechanism [De Loecker et al. \(2016\)](#) identified in India’s 1990s liberalisation, where tariff *cuts* raised markups because marginal costs fell faster than prices. The two theories disagree about the sign, so the sign is evidence. Consider the exchange rate: a real appreciation cheapens imports in rupee terms, and the import-discipline hypothesis predicts markup compression concentrated in import-exposed industries — a difference-in-differences prediction that survives any national confounder. Consider monopsony: in wage-posting models the markdown shrinks when labour demand tightens, so protection that props up product demand should *erode* labour-market power even as import competition strengthens it — a channel the trade-and-labour literature is only beginning to quantify. And consider growth: in economies with adjustment costs and heterogeneous entrants ([Asker et al., 2014](#)), expansion episodes mechanically widen measured revenue-productivity dispersion — growth can *raise* measured misallocation without any worsening of policy — against the older presumption that development and allocative efficiency improve together.

Stated as questions: **Q1** Does protection raise markups (Melitz–Ottaviano) or lower them (incomplete pass-through), and does the exchange rate discipline margins in exposed industries? **Q2** Does trade policy reach the labour market — the markdown and monopsony incidence — and in which direction? **Q3** What moves misallocation: the border (tariffs, the rupee) or domestic conditions (state growth, labour-market structure, geography)? A fourth question is forced on the analysis by the answers to the first three: **Q4** does causation also run backward — does Indian tariff-setting respond to industry conditions? — because if it does, naive regressions of outcomes on tariffs are contaminated

by policy endogeneity, and the identification strategy must be built to survive it.

The empirical challenges, and the design they dictate, are four. **(i) National series are collinear with time.** The REER, GDP growth, inflation, and the repo rate do not vary across industries within a year, so no within estimator with year effects can identify their levels. The workhorse design is therefore the exposure (shift-share) tradition of the trade literature (Topalova, 2010; Topalova and Khandelwal, 2011; De Loecker et al., 2016): industry-varying tariffs enter directly; the national exchange rate enters interacted with each industry’s *predetermined* (base-year 2012) share of manufacturing imports, so identification comes from the differential response of exposed versus sheltered industries, with year effects absorbing every purely national shock. **(ii) The tariff data do not exist at the industry classification of the outcomes.** Applied tariffs live on HS-6 trade lines; the outcomes live on NIC-2008 industries. We build the bridge ourselves from the official United Nations correspondence tables (HS2012↔CPC 2.1, HS2017↔CPC 2.1, CPC 2.1↔ISIC Rev. 4, which NIC-2008 adapts), in the tradition of Debroy and Sathanam (1993) — 4,847 manufacturing HS-6 lines mapped, validated line by line on known industries (Appendix A). **(iii) Policy is endogenous.** We address it three ways: predetermined exposure weights; a lag specification; and, frontally, a panel vector autoregression that tests the feedback directly and finds it (Section 5.3). **(iv) Eleven years.** The panel VAR must extract dynamics from $T = 11$; we select the lag order by information criteria, demean by year to remove the decade’s common trends after documenting that the levels system is explosive, and read impulse responses as short-horizon evidence, not structure.

The findings, previewed in the abstract, resolve the theoretical ambiguities cleanly. Protection does not buy power (every tariff–markup coefficient negative; dynamic responses negative for one to four years) — the De Loecker et al. mechanism in reverse. The exchange rate disciplines (appreciation compresses markups in exposed industries at both industry levels). Trade policy does reach the labour market at the finer industry level, with the theoretically expected sign (protection erodes monopsony; import competition feeds it). Misallocation is orthogonal to trade policy in every specification but rises within states with development — the transition signature. And tariff-setting itself responded to industry conditions (markups and misallocation Granger-cause tariffs), which both vindicates the identification design and reframes the post-2018 protection episode as endogenous policy.

The paper contributes to four literatures. To the *trade-and-markups* literature it adds the first industry-panel evidence from India’s *re-protection* decade, closing the loop opened by the liberalisation studies (Krishna and Mitra, 1998; De Loecker et al., 2016): the same incomplete-pass-through economics, running in the opposite policy direction, produces the mirror-image sign. To the *trade-and-labour-power* literature it contributes an industry-level estimate of protection’s effect on monopsony incidence — to our knowledge

the first for a developing economy — with a coherent demand-channel interpretation. To the *misallocation* literature it adds determinants: the robust non-result on trade and the robust positive association with state development discipline what theories of Indian misallocation must explain (domestic wedges, transition dynamics — not the border). To the *economic geography* of development it adds a five-layer state atlas of market power and misallocation and the agglomeration gradient beneath it: every margin improves with market thickness.

Table 1 collects the headline estimates. Section 2 situates the paper; Section 3 documents the data construction, variable by variable; Section 4 develops the two estimating frameworks equation by equation; Sections 5–6 present results; Section 7 interprets; Section 8 concludes. Appendices detail the concordance (A), the tariff panel (B), the panel-VAR diagnostics (C), and the complete 72-specification robustness grid (E).

Table 1: Findings at a glance. Outcomes: median markup μ , median markdown ψ , monopsony incidence, $\text{var}(\log \text{TFPR}) \times 100$. P-values clustered by panel unit.

Object	Estimate
Tariff $\rightarrow$ markup (NIC-2 baseline)	-0.0048 ($p = .086$); all 10 industry specs negative
import-weighted / no-COVID / nat. controls	-0.0028 (.010) / -0.0050 (.050) / -0.0063 (.021)
REER $\times$ exposure $\rightarrow$ markup (NIC-2 / NIC-3)	-0.039 (.038) / -0.060 (.050)
Tariff $\rightarrow$ monopsony incidence (NIC-3)	-0.0019 (.067)
Tariff $\rightarrow$ misallocation	null everywhere (min $p = .38$ across 12 specs)
ln GSDP $\rightarrow$ misallocation (state)	$+39.1$ (.030); no-COVID $+44.4$ (.009)
Granger: markup $\rightarrow$ tariff / misalloc. $\rightarrow$ tariff	$p = .016$ / $p = .050$ (NIC-3)
pVAR stability (levels / year-demeaned)	max modulus 1.107 / 0.713
Local labour-market HHI (emp.-wtd mean)	0.374 (2012) $\rightarrow$ 0.343 (2022)
Agglomeration gradient (corr. with ln state emp.)	$\mu -0.30$; $\psi -0.28$; mono -0.26 ; misalloc. -0.22
Rural–urban monopsony gap	$+5.8\text{pp}$ pooled; positive in every year

2 Related literature

Trade policy and markups. The theoretical ambiguity is genuine. In monopolistic competition with variable elasticities, protection that shields incumbents raises margins; in the marginal-cost channel, tariffs on inputs raise costs, and with incomplete pass-through measured markups fall. For India the liberalisation evidence is unusually rich: [Krishna and Mitra \(1998\)](#) document pro-competitive discipline and productivity growth after the 1991 opening; [Topalova and Khandelwal \(2011\)](#) find liberalisation raised firm productivity through both competition and cheaper inputs; and [De Loecker et al. \(2016\)](#), with product-level prices, deliver the landmark decomposition — tariff cuts *raised* markups because marginal costs fell by more than prices. Our decade supplies the reverse policy experiment: after 2018 India’s applied tariffs turned up for the first time in a generation,

and the incomplete-pass-through logic predicts exactly the negative tariff–markup association we estimate. **Exchange rates and discipline.** The import-discipline hypothesis — foreign price pressure compresses domestic margins — has its cleanest modern implementations in exposure designs where a national price (here the REER) interacts with predetermined industry exposure; our implementation follows the shift-share tradition (Topalova, 2010; De Loecker et al., 2016) with base-2012 import shares as the exposure. **Trade and labour-market power.** That import competition can strengthen employer wage-setting power (by slackening labour demand) while protection erodes it is implicit in monopsony theory (Manning, 2003) but rarely estimated; our markdown and incidence results provide industry-level evidence. **Determinants of misallocation.** The measurement literature (companion paper 2; Hsieh and Klenow, 2009) is largely silent on time-varying determinants; Asker et al. (2014) supply the transition mechanism through which growth raises measured dispersion; Hsieh et al. (2026) supply the political-economy rationale for persistence — removing distortions can be distributionally unattractive — which our endogenous-tariff finding echoes on the trade margin. **Spatial economics of market power.** Labour-market concentration work (Azar et al., 2022) and the agglomeration tradition (Ellison and Glaeser, 1997) frame the spatial layer: local labour markets as state×industry cells, agglomeration as the state employment distribution, and the prediction — borne out — that thickness disciplines every margin of market power.

3 Data: five layers, each documented

The unit-level foundation is the cleaned ASI factory panel of the companion papers: 423,127 plant-years on 142,493 factories, 2012–13 to 2022–23, NIC-2008 divisions 10–33. This paper consumes the plant-level *estimates* — the translog markup μ_{it} , the markdown ψ_{it} ($\psi > 1$ = monopsony), and within-industry TFPR deviations — and aggregates them into three outcome panels: NIC-2 × year (253 cells), NIC-3 × year (748 cells, 69 groups), and state × year (377 cells). Aggregation uses medians for μ and ψ (robust to the heavy right tails documented in paper 1), the share of plants with $\psi > 1$ for monopsony incidence (excluding divisions 18 and 30 on the data-quality grounds established there), and the within-NIC-4×year variance of log TFPR (×100) for misallocation, exactly as in paper 2. To these are merged five determinant layers.

Because the dependent variables are themselves *estimates*, three outcome-measurement choices deserve their justification. Medians, not means, carry μ and ψ into the panels: the plant-level distributions are heavily right-skewed (paper 1), and a cell mean would let a handful of extreme plants move an industry-year observation — the median makes the panel robust to exactly the tails that winsorisation handles at plant level. Divisions 18 and 30 are excluded from markdown-based outcomes because their reported factor shares imply negative value added for a seventh to a third of plants (a job-work reporting arte-

fact documented in paper 1), which would poison the numerator of any aggregate built on them. And the econometric consequence of estimated outcomes is attenuated *precision*, not bias: sampling error in a cell median inflates the residual variance of (4) and widens the clustered standard errors, but it does not correlate with the tariff or exchange-rate regressors, so the coefficients are unbiased and the reported p -values conservative.

3.1 The tariff panel: WITS lines to NIC industries

India’s effectively-applied (AHS) tariffs at the HS-6 level, 2011–2023, were extracted from the WITS database: 85,793 line-years, with MFN rates as fallback and robustness. The mapping from trade lines to industries is built from the official UN correspondence tables — HS2012↔CPC 2.1 and HS2017↔CPC 2.1, chained through CPC 2.1↔ISIC Rev. 4, which NIC-2008 adapts at the 4-digit level — covering 4,847 manufacturing HS-6 lines. The concordance was validated industry by industry on unambiguous cases (knitted apparel to division 14; pharmaceuticals to 21; primary-form plastics to 20 against plastic *articles* to 22; and so on); Appendix A documents the chain and its two vintage regimes. The headline industry tariff is the simple average over HS lines,

$$\tau_{jt} = \frac{1}{n_j} \sum_{h \in j} \text{AHS}_{ht}, \quad (1)$$

deliberately *unweighted*: import-weighted averages re-weight lines by import volumes that themselves respond to tariffs (a high tariff that kills imports of its line drops out of the weighted mean — survivorship in the regressor), so the weighted version is the robustness check, not the baseline. The resulting panels are complete: 23 divisions and 68 groups over 13 years. The 2022 cross-section illustrates the policy structure the panel exploits (Appendix Table 10): beverages (110 per cent), motor vehicles (40), tobacco (33) and food products (28) remain heavily protected; basic metals (5.5), electronics (6.2), machinery (6.5) and chemicals (6.6) face low tariffs.

3.2 Exchange rates and the exposure interaction

The nominal and real effective exchange rates are the BIS broad indices (base 2020 = 100), averaged from monthly observations over the April–March fiscal year to align with ASI accounting years. The real index traces a sharp depreciation into 2013–14 (the taper-tantrum year, 86.9), a five-year appreciation to a 2017–18 peak (102.0), and a plateau near 100; the nominal index falls monotonically from 117.6 to 97.0, the wedge reflecting India’s inflation differential. Because the REER is national, its level is absorbed by year effects; its identifying content is the interaction

$$\text{reer.exp}_{jt} = (\text{REER}_t - 100) \times s_j^{\text{imp}, 2012}, \quad (2)$$

where $s_j^{\text{imp},2012}$ is industry j 's *predetermined* base-year share of total manufacturing imports. Predetermination is the point: had the exposure been measured contemporaneously, industries whose imports responded to the rupee would shift their own weights, contaminating the design; frozen 2012 shares make (2) a textbook shift-share regressor whose identification comes solely from the differential response of exposed versus sheltered industries to a common national price.

3.3 State development, labour-market concentration, and geography

State real GSDP (constant 2011–12 prices) comes from the MOSPI state accounts (per-capita NSDP and growth rates as robustness). A *local labour market* is a state $\times$ NIC-3 $\times$ year cell, its concentration the employment Herfindahl

$$\text{HHI}_{mt} = \sum_{i \in m} s_{imt}^2, \quad (3)$$

with s_{imt} plant i 's employment share of the cell; employment-weighted means aggregate to the national series, and the state $\times$ year mean enters the state model. Two caveats carry through: cells under five plants are flagged thin, and ASI coverage of the organised sector alone means measured concentration understates a worker's true outside options only to the extent unorganised employers compete for the same workers. Geography uses the plant's state (characters 6–7 of the factory identifier, verified against the official layout) and the Block-A rural/urban flag, extracted for all eleven rounds and merging to 100 per cent of estimation plant-years.

3.4 The macroeconomic backdrop

Table 2 assembles the environment: a decade of falling then recovering trade openness (55.8 to 37.8 to 50.7 per cent of GDP), halving inflation, the COVID collapse and rebound (−5.8, +9.7 per cent), the REER cycle, and — the policy reversal at the paper's centre — mean applied tariffs falling from 17.1 (2012) to 15.1 (2016) and climbing back to 17.0 by 2022.

4 Empirical framework

4.1 The static workhorse: two-way fixed effects with exposure

For the industry panels the estimating equation is

$$Y_{jt} = \beta_1 \tau_{jt} + \beta_2 \text{reer_exp}_{jt} + \alpha_j + \delta_t + \varepsilon_{jt}, \quad (4)$$

Table 2: The macroeconomic environment, 2012–2022. Trade, growth, inflation, manufacturing share: World Bank WDI. REER/NEER: BIS broad indices (2020=100), fiscal-year averages. Tariff: unweighted mean over NIC-2 divisions of the applied (AHS) simple-average tariff of Section 3.1.

Year	Trade (%GDP)	Growth (%)	CPI (%)	Manuf. (%GDP)	REER	NEER	Tariff (%)
2012	55.8	5.5	9.5	15.8	91.6	117.6	17.1
2013	53.8	6.4	10.0	15.3	86.9	106.4	17.1
2014	48.9	7.4	6.7	15.1	92.5	108.6	16.8
2015	41.9	8.0	4.9	15.6	96.6	109.6	15.9
2016	40.1	8.3	5.0	15.2	98.7	108.9	15.1
2017	40.7	6.8	3.3	15.0	102.0	110.6	15.8
2018	43.6	6.5	3.9	14.9	96.9	103.8	15.8
2019	39.9	3.9	3.7	13.5	100.5	104.8	17.0
2020	37.8	−5.8	6.6	14.1	99.6	98.6	16.4
2021	45.4	9.7	5.1	14.4	100.3	97.0	17.1
2022	50.7	7.6	6.7	13.3	100.8	97.0	17.0

where Y is one of the four outcomes, α_j and δ_t are industry and year effects, and standard errors are clustered by industry. Each term earns its place. The industry effects α_j absorb everything permanent about an industry — its technology, its typical markup, its baseline protection — so β_1 is identified from *changes* in an industry’s tariff relative to other industries’ changes; permanently protected, permanently high-markup industries contribute nothing. The year effects δ_t absorb every national shock — GDP growth, inflation, the repo rate, the REER level, demonetisation, GST, COVID — which is what makes the surviving variation credibly policy-specific; the price of that credibility is that no purely national series can appear in levels, which is why the REER enters only through (2) and why the robustness grid includes a specification replacing δ_t with observed national controls (REER, growth, inflation) as a check that the year effects are not doing something stranger than intended. Clustering by industry allows arbitrary serial correlation within an industry’s eleven observations — with $T = 11$ and serially persistent tariffs, failure to cluster would overstate precision badly. For the state panel the trade pair is replaced by $\ln$ real GSDP and the local labour-market HHI, with state and year effects and clustering by state.

A word on what the shift-share design does and does not assume, in light of the modern econometric literature on such regressors (Goldsmith-Pinkham et al., 2020). Identification of β_2 requires that the base-year import shares be uncorrelated with industry-specific *trends* in the outcome that would have occurred without exchange-rate variation — the “exogenous shares” reading. Three features support it here: the shares are frozen four years before the REER’s main appreciation episode; industry effects absorb the level of exposure entirely, so only the interaction identifies; and the placebo margin is built in — if exposed industries merely trended differently, the same interaction should “move”

outcomes that theory says it should not, and it does not (misallocation, Table 3). What (4) cannot deliver, by construction, is dynamics (does a tariff shock’s effect build or fade?) and feedback (do tariffs respond to outcomes?); for those the design turns to the panel VAR.

4.2 Dynamics and feedback: the panel VAR

The dynamic system stacks the policy variable and the three outcomes,

$$\mathbf{y}_{jt} = (\tau_{jt}, \mu_{jt}, \psi_{jt}, V_{jt})', \quad \mathbf{y}_{jt} = \mathbf{A} \mathbf{y}_{jt-1} + \mathbf{u}_j + \mathbf{e}_{jt}, \quad (5)$$

with V the misallocation measure and $\mathbf{u}_j$ industry fixed effects. Estimating (5) by within-OLS would be biased: with $T = 11$, demeaning induces correlation between the transformed lag and the transformed error (Nickell bias). The estimator is therefore GMM on *forward orthogonal deviations* (the Helmert transformation),

$$\tilde{e}_{jt} = c_t \left(e_{jt} - \frac{1}{T-t} \sum_{s>t} e_{js} \right), \quad c_t = \sqrt{\frac{T-t}{T-t+1}}, \quad (6)$$

which subtracts from each observation the mean of its *future* values rather than the overall mean: fixed effects are removed, the scaling c_t preserves the error variance, and — the point of the construction — past values remain valid instruments, because only future information is used in the transform. Lags one to three instrument the system, following [Abrigo and Love \(2016\)](#). The primary panel is NIC-3 (69 groups; more cross-sectional units is what GMM needs), NIC-2 as the small- N check. Lag order one is selected by the [Andrews and Lu \(2001\)](#) criteria (MBIC -112.2 at lag 1 versus -42.4 at lag 2).

Two specification decisions matter for interpretation and are documented rather than buried. **First, the levels system is explosive:** its maximum eigenvalue modulus is 1.107, violating stability — the decade’s common secular trends (declining misallocation, the post-2018 tariff drift) make shocks appear permanent. Following [Love and Zicchino \(2006\)](#), all variables are demeaned by year (removing common time effects, the VAR analogue of δ_t) before estimation; the transformed system is comfortably stable (maximum modulus 0.713 at NIC-3, 0.740 at NIC-2) and all impulse responses die out (Appendix C). **Second, orthogonalised impulse responses require an ordering.** The Cholesky ordering places the tariff first — policy is administratively set and the slowest-moving variable, so it may affect outcomes contemporaneously but respond to them only with a lag — then the markup, the markdown, and misallocation last, as the most endogenous outcome object. Confidence bands use 200 Monte Carlo draws from the estimated coefficient distribution. Granger-causality tests, which do not depend on the ordering, carry the feedback results.

5 Results: the industry panels

5.1 Aggregate previews

The raw series already hint at the answer. The markup decline documented in paper 1 was *steepest* in the rising-tariff years after 2018; the 2013–2017 real appreciation coincided with the fastest markup compression in import-exposed industries; and monopsony incidence declined secularly (rural 49 to 44 per cent, urban 44 to 39) — a national trend the year effects absorb in everything that follows. Aggregate correlations cannot separate common trends from causation; the panel evidence can.

5.2 Static estimates

Table 3 reports (4) for all four outcomes on both industry panels and the state panel; Table 4 condenses the robustness grid (complete grid: Appendix E).

Table 3: Macroeconomic determinants: two-way fixed-effects estimates. Each cell reports the coefficient with cluster-robust p -value in parentheses. Industry panels: applied tariff and REER $\times$ exposure. State panel: ln real GSDP and labour-market HHI.

Panel	Regressor	med μ	med ψ	Monopsony	var TFPR
NIC-2	Applied tariff	-0.0048* (0.086)	0.0009 (0.757)	0.0003 (0.853)	-0.0001 (0.999)
NIC-2	REER $\times$ exposure	-0.0391** (0.038)	0.0177 (0.746)	0.0109 (0.520)	0.0720 (0.974)
State	ln real GSDP	0.0362 (0.562)	-0.0513 (0.681)	-0.0824 (0.236)	39.0816** (0.030)
State	Labour-market HHI	-0.0096 (0.950)	-0.4438* (0.083)	0.0674 (0.523)	33.2314 (0.471)

Table 4: Robustness of the two headline results (key coefficient, cluster-robust p -value). Left: markup on tariff. Right: monopsony incidence on tariff, NIC-3. “Nat. controls” replaces year effects with the fiscal-year REER, GDP growth and CPI inflation.

Specification	μ on τ , NIC-2	μ on τ , NIC-3	Mono. on τ , NIC-3
Baseline (TWFE)	-0.0048* (0.086)	-0.0021 (0.206)	-0.0019* (0.067)
One-year lag	-0.0040* (0.086)	-0.0018 (0.271)	-0.0018* (0.069)
Import-weighted tariff	-0.0028** (0.010)	-0.0016 (0.108)	-0.0010 (0.133)
Excluding 2020–21	-0.0050** (0.050)	-0.0021 (0.114)	-0.0014 (0.145)
Unit FE + national controls	-0.0063** (0.021)	-0.0031* (0.062)	-0.0017* (0.073)

Result 1: protection does not raise markups; if anything it lowers them. The tariff coefficient on the median markup is negative in *all ten* industry-panel specifications (five per panel: baseline, lag, import-weighted, excluding COVID, national controls). At NIC-2 the baseline is -0.0048 ($p = .086$), strengthening — not weakening — under scrutiny: import-weighted -0.0028 ($p = .010$), excluding 2020–21 -0.0050 ($p = .050$), national controls -0.0063 ($p = .021$), one-year lag -0.0040 ($p = .086$). Quantitatively a ten-point tariff increase is associated with a median markup lower by 0.03–0.06 — two

to five per cent of the markup level, small in economic magnitude but decisive in sign, because the two theories of protection disagree precisely about the sign. The reading that fits both this result and the 1990s evidence is incomplete pass-through of the cost channel: tariffs raise input costs (much of Indian manufacturing imports intermediates), costs pass into prices less than one-for-one, and measured margins compress — [De Loecker et al. \(2016\)](#) run in reverse.

Result 2: the exchange rate disciplines exposed industries. The shift-share coefficient (2) on the markup is -0.039 ($p = .038$) at NIC-2 and -0.060 ($p = .050$) at NIC-3. Because it is identified within industry *and* within year, this is the cleanest quasi-experimental statement in the static design: when the rupee strengthens, industries that import more see their markups squeezed relative to sheltered industries. Import discipline has migrated from the tariff line (where variation is now a few points around a low mean) to the exchange rate (where the decade’s swings were large).

Magnitudes against observed policy swings. The coefficients acquire economic content when scaled by the variation the decade actually supplied. The post-2016 tariff reversal raised the mean applied tariff by 1.9 points (15.1 to 17.0, Table 2); at the baseline and national-controls coefficients this maps to a median markup lower by 0.009–0.012 — protection “cost” its industries roughly one per cent of their margin, the opposite of its presumed purpose, though modest in size. The REER swing was far larger: from the 2013 trough (86.9) to the 2017 peak (102.0) is 15.1 index points, so an industry carrying a ten per cent share of manufacturing imports is estimated to have compressed its markup by $0.039 \times 15.1 \times 0.10 \approx 0.06$ relative to an unexposed industry — five per cent of the markup level, concentrated in exactly the years the aggregate markup series compressed fastest. The exchange rate, not the tariff schedule, supplied the decade’s effective dose of import discipline.

Result 3: trade policy reaches the labour market at the finer industry level. Markdowns are unresponsive at NIC-2, but the NIC-3 panel — 69 groups, ten times the effective cross-section — detects protection *eroding* monopsony: -0.0019 ($p = .067$), confirmed with lags ($p = .069$) and national controls ($p = .073$), while appreciation in exposed industries weakly *raises* monopsony incidence ($p = .095$). The coherent reading runs through product demand: protection props up demand for the industry’s output, tightening its labour demand and eroding wage-setting power; import competition slackens it and does the opposite. The labour-market face of trade policy is the mirror image of its product-market face — and it is the *pro-worker* face of protection, a margin absent from the Indian debate.

Result 4: misallocation is orthogonal to trade policy. Across every specification — both panels, all five variants, tariff and REER-exposure alike — the coefficient on TFPR dispersion is tiny and insignificant (nothing below $p = .38$; NIC-2 baseline: -0.0001 , $p = .999$). This non-result is sharp, replicable, and informative: whatever

moved Indian misallocation in this decade, it was not the border. Set against [Topalova and Khandelwal \(2011\)](#), where 1990s liberalisation moved productivity and allocation, the contrast says the low-hanging allocative fruit of trade reform had been picked; the remaining wedges (paper 2: capital-market above all) are domestic.

5.3 Dynamics and feedback: panel-VAR results

Figure 1 displays the orthogonalised impulse responses of the year-demeaned NIC-3 system; Table 5 the Granger tests.

Table 5: Panel-VAR Granger-causality tests (year-demeaned system, GMM/forward orthogonal deviations, lag 1; instruments lags 1–3; stability satisfied, max modulus 0.713 / 0.740). Selected hypotheses; full tables in the estimation logs.

Null: row does not Granger-cause column	NIC-3 χ^2 (p)	NIC-2 χ^2 (p)
Markup $\nrightarrow$ Tariff	5.82 (0.016)	2.81 (0.094)
Misallocation $\nrightarrow$ Tariff	3.84 (0.050)	0.49 (0.482)
Tariff $\nrightarrow$ Markdown	3.37 (0.067)	0.02 (0.881)
Tariff $\nrightarrow$ Misallocation	3.05 (0.081)	0.46 (0.497)
Markdown $\nrightarrow$ Markup	3.54 (0.060)	6.88 (0.009)
Markup $\nrightarrow$ Markdown	1.37 (0.242)	11.40 (0.001)

Before the findings, a note on how to read the two objects. An orthogonalised impulse response answers: after a one-standard-deviation innovation to the shocked variable — purged, through the Cholesky ordering, of its contemporaneous correlation with variables earlier in the ordering — how does each variable in the system evolve over the following years, relative to its no-shock path? A Granger test answers a weaker but ordering-free question: do lagged values of one variable improve the prediction of another beyond the latter’s own history? Neither is structural causation; together, with the sign pattern of the static design, they triangulate.

Three dynamic findings. **First, the static signs survive as dynamics:** a tariff shock lowers the markup over the following one to four years and lowers the markdown for one to two — the impulse-response counterpart of Results 1 and 3 — while temporarily *raising* misallocation for two to three years before dying out (Granger $p = .081$): discrete protection episodes cause short-lived allocative disruption even though there is no level relationship. The three horizons compose into one narrative: a protection episode reshuffles demand toward protected lines, which lands unevenly on heterogeneous plants (the transitory misallocation hump); imported-input costs pass into the cost base and compress measured margins over the longer window; and the labour market, whose wage-setting responds to current demand conditions, adjusts over the shortest one. **Second, and more consequential: the feedback runs backward, and it is the stronger direction.** Markups Granger-cause tariffs ($p = .016$) and misallocation Granger-causes

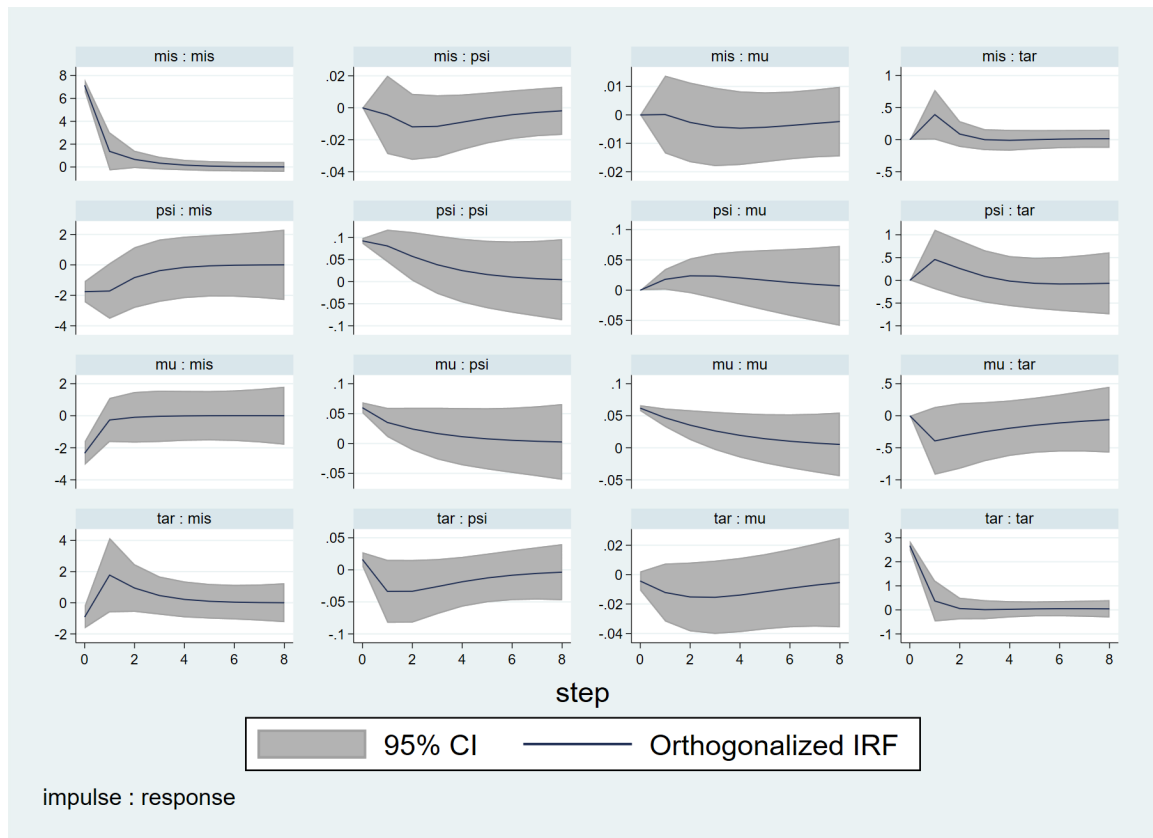


Figure 1: Orthogonalised impulse responses, year-demeaned NIC-3 panel VAR (Cholesky order: tariff → markup → markdown → misallocation; 95 per cent Monte Carlo bands, 200 draws).

tariffs ($p = .050$): Indian tariff-setting over this decade *responded* to industry conditions — falling after markups rose, rising where misallocation (a proxy for industry distress and heterogeneity) grew. This political-economy feedback vindicates the predetermined-exposure identification of the static design and reframes the post-2018 protection episode: it was not an exogenous experiment but endogenous policy, which any evaluation must model as such. It also echoes, on the trade margin, the persistence logic of [Hsieh et al. \(2026\)](#): distortions persist when their removal is politically or distributionally unattractive. **Third, the two margins of market power are dynamically joint:** at NIC-2, markups and markdowns Granger-cause each other ($p = .001, .009$), confirming at industry frequency the plant-level joint determination documented in paper 1 — another reason the two margins must be estimated together, never one at a time.

5.4 The state panel: growth and misallocation

The state estimates (Table 3, bottom panel) deliver one null with content and one robust surprise. The null: median markups are unrelated to state development ($+0.036$, $p = .562$) — consistent with paper 1’s geography, the product-market margin is set in national markets and state macroeconomics does not reach it. The markdown margin shows suggestive movement — it falls with local labour-market concentration (-0.44 , $p = .083$; a thin-market composition effect unpacked in Section 5.5) and with lagged GSDP (-0.195 , $p = .093$), hinting that state growth erodes wage-setting power with a delay, consistent with the thickening-markets interpretation of Section 6.

The surprise: **within states over time, growth is associated with rising misallocation.** The coefficient of $\ln$ real GSDP on TFPR dispersion is $+39.1$ ($p = .030$); excluding COVID years it *strengthens* to $+44.4$ ($p = .009$); per-capita NSDP gives $+37.7$ ($p = .065$); the lag $+30.1$ ($p = .111$). Growth *rates* carry nothing ($p = .78$) — it is the development level, trending within states, that does. Since state and year effects absorb permanent differences and the national decline, the estimate says: states growing faster than the national trend saw within-industry dispersion fall by *less*. The interpretation we develop in Section 7 is Schumpeterian rather than pathological — growth episodes draw in heterogeneous entrants and expand constrained incumbents unevenly, widening measured dispersion in transition ([Asker et al., 2014](#)) — and it is flagged as an association: with eleven years, the alternative (state growth generating distortions) cannot be excluded.

5.5 Labour-market concentration

Table 6 reports the national concentration series.

Concentration is high but falling: the employment-weighted mean HHI declines monotonically from 0.374 to 0.343 (median market 0.266 to 0.228) while plants per market rise from 24 to 29 — Indian local labour markets *thickened* all decade, in step with the declin-

Table 6: Local labour-market concentration, 2012–2022. Market = state $\times$ NIC-3 $\times$ year cell; HHI per (3).

Year	Emp.-weighted mean HHI	Median market HHI	Mean plants/market
2012	0.374	0.266	24.3
2013	0.366	0.265	25.3
2014	0.361	0.256	26.5
2015	0.370	0.260	25.9
2016	0.356	0.259	27.8
2017	0.358	0.245	27.4
2018	0.361	0.248	27.2
2019	0.362	0.254	27.0
2020	0.351	0.237	28.3
2021	0.352	0.237	28.8
2022	0.343	0.228	29.5

ing monopsony incidence of paper 1. By the U.S. benchmarks of [Azar et al. \(2022\)](#) these levels remain deeply concentrated (an HHI of 0.34 is roughly three equal employers per market). The linkage to outcomes, however, is subtler than a concentration $\rightarrow$ monopsony arrow: in the state panel the median markdown *falls* with concentration, and across states concentration correlates at -0.91 with employment size — concentration in India is primarily a *thinness* phenomenon, proxying for small state-industries whose median plant is small with a low measured markdown even as monopsony *incidence* among its plants is high. The HHI is therefore read as a structural descriptor, the plant-level incidence as the market-power content, and the rural–urban split below as the informative spatial lens.

6 Welfare and space

6.1 The welfare-cost bounds over time

Table 7 updates the static Harberger bounds of paper 1 (half the wedge times distorted expenditure, plant by plant) as an annual series.

The Lerner index ends the decade below its start (0.143 vs 0.160) — aggregate product-market power never recovered its 2012 level — yet the product-side DWL bound drifts *up*, from 5.8 to 9.9 per cent of sales (peaking 12.2 in 2019), because the bound is convex in the wedge and markup *dispersion* widened. The labour triangle is smaller and stable (2.2–2.8 per cent of the wage bill). The ranking — misallocation (paper 2) $\gg$ markup dispersion $>$ markdown triangle — is the welfare bottom line of the series, and the flat-Lerner-rising-DWL divergence is itself a dispersion fact: the harm of market power operates through heterogeneity, not the average wedge.

Table 7: Static welfare bounds by year: sales-weighted Lerner index, product-market dead-weight loss (per cent of sales), labour-market (markdown) loss (per cent of the wage bill).

Year	Lerner	DWL _p (% sales)	DWL _L (% wage bill)
2012	0.160	5.8	2.6
2013	0.161	7.5	2.5
2014	0.157	7.8	2.3
2015	0.128	7.7	2.3
2016	0.123	9.4	2.2
2017	0.134	8.5	2.3
2018	0.139	8.7	2.4
2019	0.139	12.2	2.3
2020	0.133	8.9	2.3
2021	0.148	7.3	2.6
2022	0.143	9.9	2.8

6.2 Agglomeration and the spatial gradient

Indian organised manufacturing is heavily agglomerated: the employment Herfindahl across states is 0.077, and the top five states — Tamil Nadu (15.1 per cent), Maharashtra (13.0), Gujarat (10.6), Karnataka (7.4), Uttar Pradesh (7.0) — host 53 per cent of factory employment. Industries differ sharply (Table 8): tobacco is the extreme (spatial HHI 0.565, top state 74 per cent), followed by leather (0.215) and apparel (0.203) — resource-tied and labour-pool industries — while intermediates (chemicals 0.128, machinery 0.130) disperse toward demand, the classic [Ellison and Glaeser \(1997\)](#) pattern.

The agglomeration gradient summarises the spatial equilibrium: the plant-weighted correlation of log state manufacturing employment is -0.30 with the median markup, -0.28 with the median markdown, -0.26 with monopsony incidence, -0.22 with misallocation, and -0.91 with labour-market concentration. *Every margin improves with market thickness*: agglomerated states are more competitive in product markets, less monopsonistic in labour markets, and better allocated. Thickness is the single most consistent correlate of competition in the data — stronger than any policy variable in the determinant models.

6.3 The rural–urban margin

The sharpest within-state contrast is rural–urban. Nationally 44.0 per cent of plant-years are rural; rural plants are monopsonistic in 44.2 per cent of cases against 38.4 urban — a gap present in every year — while median markups (1.217 vs 1.233) and misallocation (64.4 vs 63.1) barely differ across the boundary. Labour-market power, not product-market power or allocative efficiency, is what distinguishes rural from urban industrial India. The state pattern sharpens the mechanism (Figure 2): the rural excess is largest exactly where labour markets are thin and concentrated — West Bengal (+20.8

Table 8: Spatial concentration of employment by industry: Herfindahl across states and top-state share, pooled 2012–2022.

Industry (NIC-2)	Spatial HHI	Top-state share
Tobacco products (12)	0.565	0.744
Leather (15)	0.215	0.386
Wearing apparel (14)	0.203	0.313
Motor vehicles (29)	0.155	0.252
Machinery n.e.c. (28)	0.130	0.219
Chemicals (20)	0.128	0.292
Other manufacturing (32)	0.122	0.221
Furniture (31)	0.116	0.234
Other transport equipment (30)	0.110	0.184
Textiles (13)	0.108	0.203
Electronics & optical (26)	0.107	0.169
Fabricated metal (25)	0.098	0.199
Pharmaceuticals (21)	0.096	0.170
Basic metals (24)	0.096	0.186
Coke & refined petroleum (19)	0.092	0.211
Printing (18)	0.088	0.172
Rubber & plastics (22)	0.082	0.143
Food products (10)	0.079	0.155
Electrical equipment (27)	0.075	0.141
Wood products (16)	0.072	0.161
Paper (17)	0.071	0.126
Beverages (11)	0.068	0.117
Non-metallic minerals (23)	0.065	0.149

points), Uttar Pradesh (+20.5), Odisha (+16.0), Telangana (+15.8) — and it *reverses* only where the rural sector consists of large plantation-scale employers or the urban sector is itself distressed (Bihar −31.7, Assam −11.7, Punjab −5.4). The exceptions prove the mechanism: where rural labour markets are thick, or urban ones thin, the wage-setting gap closes or flips. Table 9 reports all twenty major industrial states.

Table 9: Monopsony incidence by sector and state (per cent of plants with $\psi > 1$), pooled 2012–2022, major industrial states ranked by the rural excess.

State	Rural	Urban	Gap (R–U, pp)
West Bengal	60.3	39.6	+20.8
Uttar Pradesh	58.3	37.8	+20.5
Odisha	61.7	45.7	+16.0
Telangana	51.9	36.1	+15.8
Karnataka	41.1	30.1	+11.0
Himachal Pradesh	48.6	38.2	+10.5
Maharashtra	39.0	29.7	+9.3
Jharkhand	58.9	51.4	+7.5
Madhya Pradesh	58.1	51.2	+6.9
Tamil Nadu	34.2	28.8	+5.4
Gujarat	46.4	41.4	+4.9
Chhattisgarh	66.8	62.6	+4.2
Kerala	37.2	33.3	+3.9
Haryana	39.5	36.5	+3.1
Uttarakhand	50.1	49.3	+0.7
Rajasthan	45.0	44.9	+0.1
Andhra Pradesh	50.9	52.6	−1.7
Punjab	41.1	46.5	−5.4
Assam	40.3	52.0	−11.7
Bihar	21.1	52.8	−31.7

The table’s structure is itself evidence. The gap is positive in sixteen of twenty major states and exceeds ten points in six; the four reversals are legible one by one — Bihar and Assam’s rural sectors are dominated by large plantation- and mill-scale employers whose *urban* counterparts operate in thin distressed markets, and Punjab’s dense rural agro-industrial belt is precisely a thick rural labour market. A mechanism that predicted a uniform rural excess would be suspect; one that predicts the excess *where rural markets are thin* and its reversal where they are not is what the data show.

7 Interpretation

Three theoretical frames organise the results into one account. **Trade and markups.** The negative tariff–markup association, statically and dynamically, is the incomplete-pass-through channel of [De Loecker et al. \(2016\)](#) operating in reverse: tariffs raise the cost of imported inputs; costs pass into prices less than one-for-one; measured margins

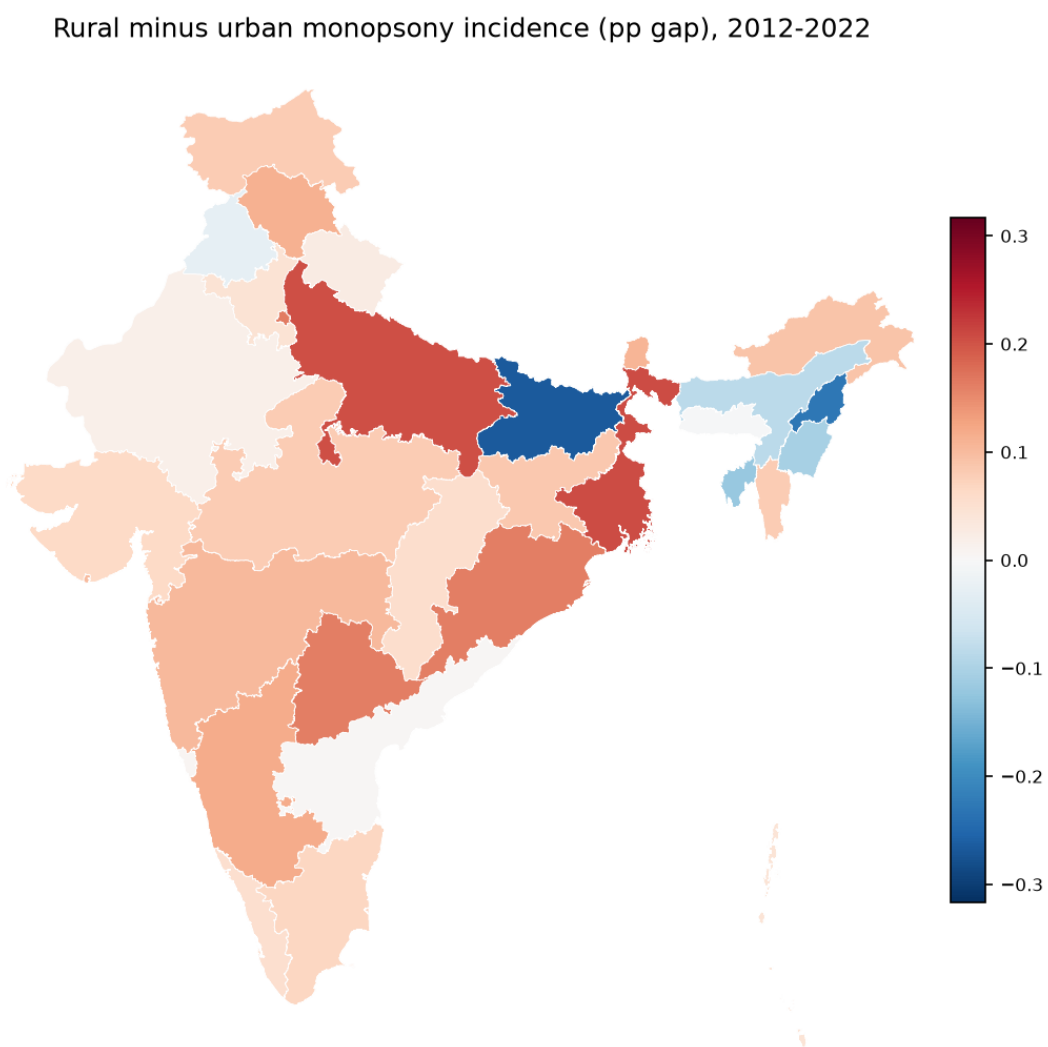
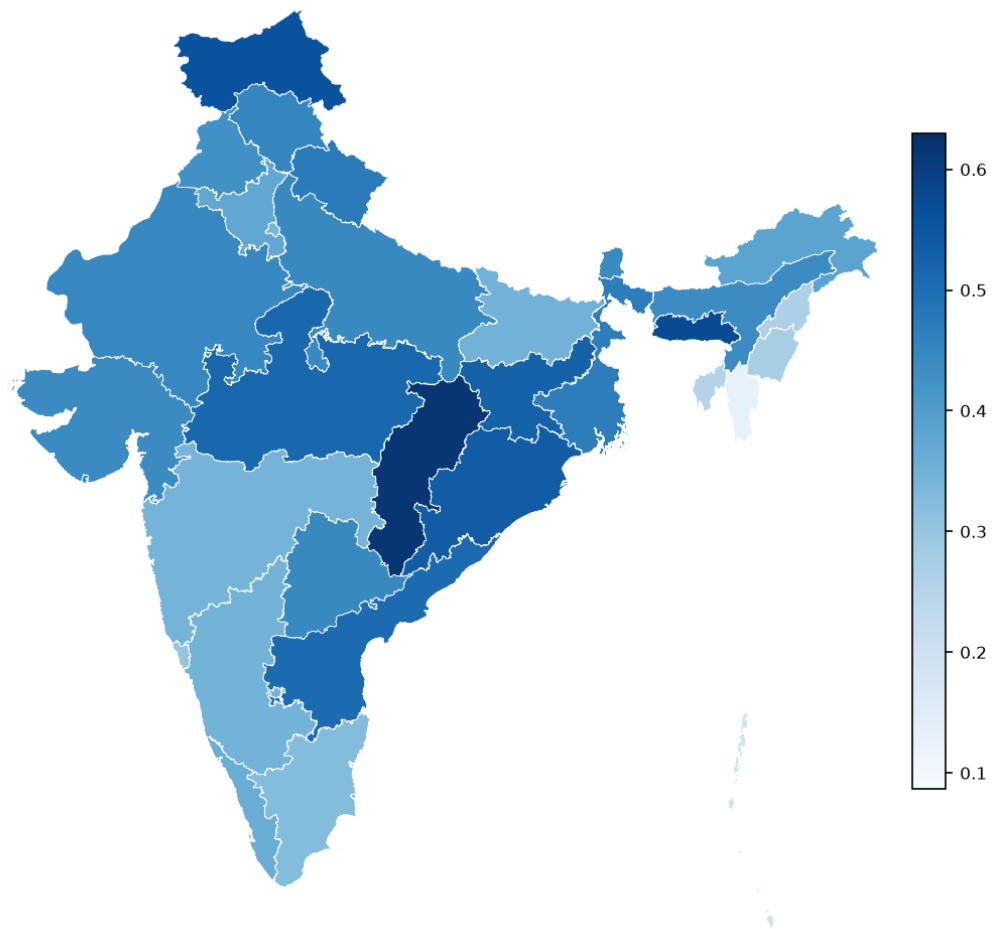
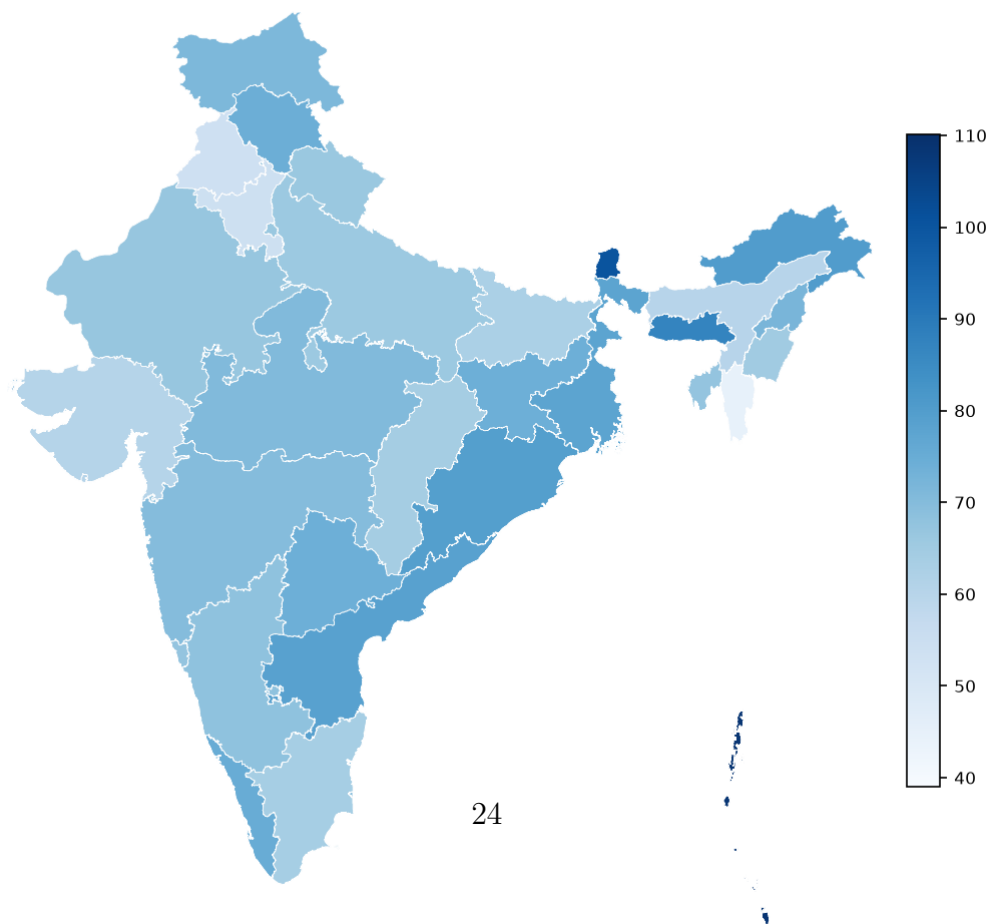


Figure 2: Rural minus urban monopsony incidence by state (percentage points), pooled 2012–2022.

Monopsony incidence by state (share with $\psi > 1$), 2012-2022



Misallocation by state ($\text{var log TFPR} \times 100$), 2012-2022



compress. The negative REER-exposure effect is the same economics run through prices rather than tariff lines: appreciation lowers rivals’ rupee prices, and exposed industries absorb part of the squeeze in margins. Together they imply that India’s post-2018 protection bought its manufacturers no pricing power — *protection without power* — while the rupee, not the tariff schedule, is now the operative lever of import discipline. **Labour-market power and product demand.** In monopsony models the markdown shrinks when labour demand tightens; protection that props up product demand therefore erodes monopsony (the NIC-3 finding), while import competition feeds it. The distributional footnote to any liberalisation argument follows: freer trade’s product-market discipline and its labour-market empowerment of employers arrive together, and the net effect on workers is an empirical question, not a presumption. **Growth and measured misallocation.** With adjustment costs and heterogeneous entrants, growth episodes widen TFPR dispersion mechanically during transitions; the within-state positive GSDP–dispersion association is exactly that signature, and it cautions against reading TFPR dispersion as a pure distortion index during growth accelerations — fast-growing states will look “worse” on this metric while doing nothing wrong. The endogenous-tariff finding, finally, supplies the political economy: policy responded to conditions, falling when margins rose and rising where distress concentrated, which is how distortions persist (Hsieh et al., 2026) and why the exposure design — not the raw regression — carries the paper’s causal weight.

The policy geography is then legible. Product-market power needs no state-level policy: it is modest, spatially uniform, disciplined by openness through the exchange rate, and unresponsive to protection. Labour-market power is a *place* problem — rural, thin-market, mineral-belt — and its documented erosion by market thickening (secularly, and with state growth at a lag) points to connectivity, mobility, and local employer density, not wage regulation alone, as the operative instruments. Misallocation is a *domestic institutions* problem — eastern-state, capital-wedge — that trade policy no longer reaches; and its transitory widening in fast-growing states is a cost of adjustment, not a failure of growth.

8 Conclusion

This paper asked what moves market power and misallocation in Indian organised manufacturing, using an eleven-year, three-level determinants panel built from plant-level estimates, a newly concorded tariff panel, exchange-rate exposure, state accounts, labour-market concentration, and a panel VAR. The answers: protection does not buy power (all ten tariff–markup coefficients negative; dynamic responses negative at one to four years); the exchange rate disciplines (appreciation compresses exposed industries’ markups at both industry levels); trade policy’s labour-market face is the mirror of its product-market face (protection erodes monopsony at the finer industry level); misallocation is

domestic (orthogonal to trade policy everywhere, rising within states with development — the transition signature); and tariff-setting is endogenous (markups and misallocation Granger-cause tariffs). Space is the sharpest determinant of labour-market power: every margin improves with agglomeration, and the rural–urban monopsony gap is the dominant within-state fact.

Four limitations bound the claims. (i) State is the finest geography available; district-level agglomeration and commuting-zone labour markets cannot be built from this extract. (ii) $T = 11$: the panel VAR is short-horizon evidence of predictive precedence, not structural causation, and is read as such. (iii) National series are absorbed by design; the paper identifies relative (cross-industry, cross-state) responses and is silent on the aggregate effect of, e.g., monetary policy on market power. (iv) The tariff measure is the output tariff; an input-tariff measure via input–output weights — the cost channel proper — and an export-side REER exposure are natural extensions. None of these caveats touches the paper’s central signs, which survive all seventy-two specifications attempted (Appendix E). The series’ bottom line stands: in the India of 2012–2022, the price of goods was disciplined by openness and indifferent to protection; the price of labour was set by the thickness of local markets; and the allocation of capital — the economy’s largest measured distortion — answered to neither, waiting on domestic reform.

References

- Abrigo, M. R. M., & Love, I. (2016). Estimation of panel vector autoregression in Stata. *The Stata Journal*, 16(3), 778–804.
- Andrews, D. W. K., & Lu, B. (2001). Consistent model and moment selection procedures for GMM estimation with application to dynamic panel data models. *Journal of Econometrics*, 101(1), 123–164.
- Asker, J., Collard-Wexler, A., & De Loecker, J. (2014). Dynamic inputs and resource (mis)allocation. *Journal of Political Economy*, 122(5), 1013–1063.
- Azar, J., Marinescu, I., & Steinbaum, M. (2022). Labor market concentration. *Journal of Human Resources*, 57(S), S167–S199.
- Brooks, W. J., Kaboski, J. P., Li, Y. A., & Qian, W. (2021). Exploitation of labor? Classical monopsony power and labor’s share. *Journal of Development Economics*, 150, 102627.
- Debroy, B., & Santhanam, A. T. (1993). Matching trade codes with industrial codes. *Foreign Trade Bulletin*, 24(1), 5–27.
- De Loecker, J., Goldberg, P. K., Khandelwal, A. K., & Pavcnik, N. (2016). Prices, markups, and trade reform. *Econometrica*, 84(2), 445–510.

- De Loecker, J., & Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471.
- Ellison, G., & Glaeser, E. L. (1997). Geographic concentration in U.S. manufacturing industries: A dartboard approach. *Journal of Political Economy*, 105(5), 889–927.
- Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8), 2586–2624.
- Hsieh, C.-T., & Klenow, P. J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4), 1403–1448.
- Hsieh, C.-T., Klenow, P. J., Kyei Manu, E., & Rockall, E. (2026). Misallocation versus inequality. Working paper.
- Krishna, P., & Mitra, D. (1998). Trade liberalization, market discipline and productivity growth: New evidence from India. *Journal of Development Economics*, 56(2), 447–462.
- Love, I., & Zicchino, L. (2006). Financial development and dynamic investment behavior: Evidence from panel VAR. *Quarterly Review of Economics and Finance*, 46(2), 190–210.
- Manning, A. (2003). *Monopsony in motion: Imperfect competition in labor markets*. Princeton University Press.
- Topalova, P. (2010). Factor immobility and regional impacts of trade liberalization: Evidence on poverty from India. *American Economic Journal: Applied Economics*, 2(4), 1–41.
- Topalova, P., & Khandelwal, A. (2011). Trade liberalization and firm productivity: The case of India. *Review of Economics and Statistics*, 93(3), 995–1009.

A The HS–NIC concordance

Applied tariffs live on HS-6 trade lines; the outcomes live on NIC-2008 industries; no official direct bridge exists. The concordance chains the official United Nations correspondence tables: HS2012↔CPC 2.1 for tariff years 2011–2016 and HS2017↔CPC 2.1 for 2017–2023 (respecting the HS revision), then CPC 2.1↔ISIC Rev. 4; NIC-2008 adapts ISIC Rev. 4, identical at the 4-digit level for manufacturing. Where an HS line maps to multiple CPC subclasses the line is assigned to each and the industry aggregate (1) averages over assigned lines; 4,847 manufacturing HS-6 lines are covered. Validation proceeded industry by industry on unambiguous anchors: knitted apparel lines land in division 14, pharmaceutical formulations in 21, primary-form plastics in 20 against plastic articles in 22, motor vehicles in 29 against parts in 29.3. The method follows the Indian tradition of trade–industry matching begun by [Debroy and Santhanam \(1993\)](#).

B The tariff panel

Table 10 reports the industry cross-section at the sample’s endpoints; the interior years are available in the archived panel (`tariff_nic2_year.csv`, 23 divisions $\times$ 13 years; a 3-digit panel with 68 groups accompanies it).

Table 10: Applied (AHS) simple-average tariffs by NIC-2 division, 2012 and 2022, with the decade change (percentage points).

Industry (NIC-2)	2012	2022	Δ
Food products (10)	32.8	28.2	-4.6
Beverages (11)	117.5	109.6	-7.9
Tobacco products (12)	32.9	33.2	+0.3
Textiles (13)	11.6	9.2	-2.3
Wearing apparel (14)	20.9	15.7	-5.2
Leather (15)	9.5	14.5	+5.0
Wood products (16)	9.7	6.8	-2.9
Paper (17)	9.2	7.5	-1.7
Printing (18)	8.8	6.9	-1.9
Coke & refined petroleum (19)	5.6	6.0	+0.3
Chemicals (20)	8.3	6.9	-1.4
Pharmaceuticals (21)	8.0	7.2	-0.9
Rubber & plastics (22)	9.9	9.3	-0.6
Non-metallic minerals (23)	9.3	9.3	-0.0
Basic metals (24)	6.3	5.7	-0.6
Fabricated metal (25)	9.7	9.0	-0.7
Electronics & optical (26)	6.9	6.7	-0.3
Electrical equipment (27)	8.6	10.5	+1.9
Machinery n.e.c. (28)	7.5	6.5	-1.0
Motor vehicles (29)	23.6	39.7	+16.2
Other transport equipment (30)	17.4	16.5	-0.9
Furniture (31)	10.0	17.0	+7.0
Other manufacturing (32)	9.4	10.1	+0.7

C Panel-VAR diagnostics

Lag selection (NIC-3 system): MBIC -112.2 at lag 1 against -42.4 at lag 2; MAIC and MQIC concur; lag 1 selected. Stability: the levels system has maximum eigenvalue modulus **1.107** — explosive, driven by the decade’s common trends in misallocation (secular decline) and tariffs (post-2018 drift). After year-demeaning (Love and Zicchino, 2006) the maximum modulus is **0.713** (NIC-3) and **0.740** (NIC-2); all roots lie strictly inside the unit circle and every impulse response decays to zero within the plotted horizon. GMM uses forward orthogonal deviations (6) with lags 1–3 as instruments (Abrigo and Love, 2016); impulse-response bands are 95 per cent Monte Carlo intervals from 200 draws of the estimated coefficient distribution. The Cholesky ordering (tariff, markup,

markdown, misallocation) is defended in Section 4.2; Granger tests (Table 5) are ordering-free.

D Reproducibility

Every estimate descends from a scripted, logged pipeline. The determinant panels (NIC-2: 253 cells; NIC-3: 748; state: 377) are built by numbered do-files from the plant-level estimate files of papers 1–2 and the external layers of Section 3; the static models, the robustness grid (Appendix E), and the panel VAR (levels, demeaned, and impulse-response runs) each save a log, and the concordance, tariff panels, exchange-rate series, state accounts, concentration series, welfare series, and spatial summaries are archived as CSV alongside the estimation .dta files. The interactive five-layer state choropleth (employment concentration, markup, markdown, monopsony incidence, misallocation) accompanies the paper as a supplementary file; the static exports appear as Figures 2–3. ASI unit records are licensed by MOSPI and cannot be redistributed; all industry- and state-level panels used here contain no unit-level information and are available from the authors, as is all code.

E The complete robustness grid

Table 11 reports every estimated specification: three panels, four outcomes, five specification variants per industry panel and five per state panel — the complete grid behind the summary claims in the text (72 estimated coefficients including the baseline REER-exposure terms). Key: *base* = two-way FE baseline; *lag* = one-year-lagged regressor; *impwtd* = import-weighted tariff; *noCOVID* = excluding 2020–21; *unitFE* = unit FE with national controls (REER, growth, inflation) in place of year effects; *growth* = GSDP growth rate; *percap* = ln per-capita NSDP.

Table 11: The complete specification grid: coefficient, cluster-robust standard error, *p*-value, observations.

Panel	Spec	Outcome / regressor	Coef.	s.e.	<i>p</i>	<i>N</i>
nic2	base	mu_med / tariff_simple	-0.0048	0.0027	0.086	253
nic2	base	mu_med / reer_exp	-0.0391	0.0177	0.038	253
nic2	lag	mu_med / L.tariff_simple	-0.0040	0.0022	0.086	230
nic2	impwtd	mu_med / tariff_wtd	-0.0028	0.0010	0.010	253
nic2	noCOVID	mu_med / tariff_simple	-0.0050	0.0024	0.050	207
nic2	unitFE	mu_med / tariff_simple	-0.0063	0.0025	0.021	253
nic2	base	psi_med / tariff_simple	0.0009	0.0029	0.757	253
nic2	base	psi_med / reer_exp	0.0177	0.0541	0.746	253
nic2	lag	psi_med / L.tariff_simple	0.0023	0.0018	0.227	230

Panel	Spec	Outcome / regressor	Coef.	s.e.	<i>p</i>	<i>N</i>
nic2	impwtd	psi_med / tariff_wtd	-0.0004	0.0010	0.713	253
nic2	noCOVID	psi_med / tariff_simple	0.0016	0.0026	0.553	207
nic2	unitFE	psi_med / tariff_simple	0.0023	0.0023	0.338	253
nic2	base	monoshare / tariff_simple	0.0003	0.0013	0.853	253
nic2	base	monoshare / reer_exp	0.0109	0.0167	0.520	253
nic2	lag	monoshare / L.tariff_simple	0.0009	0.0008	0.279	230
nic2	impwtd	monoshare / tariff_wtd	-0.0004	0.0005	0.417	253
nic2	noCOVID	monoshare / tariff_simple	0.0004	0.0013	0.779	207
nic2	unitFE	monoshare / tariff_simple	0.0004	0.0012	0.725	253
nic2	base	vartfpr100 / tariff_simple	-0.0001	0.1665	0.999	253
nic2	base	vartfpr100 / reer_exp	0.0720	2.1532	0.974	253
nic2	lag	vartfpr100 / L.tariff_simple	-0.1669	0.2492	0.510	230
nic2	impwtd	vartfpr100 / tariff_wtd	0.0251	0.0607	0.683	253
nic2	noCOVID	vartfpr100 / tariff_simple	-0.0092	0.1574	0.954	207
nic2	unitFE	vartfpr100 / tariff_simple	-0.1205	0.1362	0.386	253
nic3	base	mu_med / tariff_simple	-0.0021	0.0017	0.206	748
nic3	base	mu_med / reer_exp	-0.0595	0.0298	0.050	748
nic3	lag	mu_med / L.tariff_simple	-0.0018	0.0016	0.271	680
nic3	impwtd	mu_med / tariff_wtd	-0.0016	0.0010	0.108	739
nic3	noCOVID	mu_med / tariff_simple	-0.0021	0.0013	0.114	612
nic3	unitFE	mu_med / tariff_simple	-0.0031	0.0016	0.062	748
nic3	base	psi_med / tariff_simple	-0.0012	0.0036	0.748	748
nic3	base	psi_med / reer_exp	0.0717	0.0970	0.462	748
nic3	lag	psi_med / L.tariff_simple	-0.0037	0.0030	0.225	680
nic3	impwtd	psi_med / tariff_wtd	-0.0010	0.0021	0.656	739
nic3	noCOVID	psi_med / tariff_simple	0.0013	0.0037	0.734	612
nic3	unitFE	psi_med / tariff_simple	-0.0007	0.0033	0.830	748
nic3	base	monoshare / tariff_simple	-0.0019	0.0010	0.067	748
nic3	base	monoshare / reer_exp	0.0452	0.0267	0.095	748
nic3	lag	monoshare / L.tariff_simple	-0.0018	0.0009	0.069	680
nic3	impwtd	monoshare / tariff_wtd	-0.0010	0.0006	0.133	739
nic3	noCOVID	monoshare / tariff_simple	-0.0014	0.0009	0.145	612
nic3	unitFE	monoshare / tariff_simple	-0.0017	0.0009	0.073	748
nic3	base	vartfpr100 / tariff_simple	0.0014	0.0903	0.988	639
nic3	base	vartfpr100 / reer_exp	-0.1100	1.8688	0.953	639
nic3	lag	vartfpr100 / L.tariff_simple	-0.0795	0.0947	0.405	581
nic3	impwtd	vartfpr100 / tariff_wtd	0.0192	0.0664	0.774	639
nic3	noCOVID	vartfpr100 / tariff_simple	-0.0250	0.0808	0.758	521
nic3	unitFE	vartfpr100 / tariff_simple	-0.0286	0.0995	0.775	639
state	base	mu_med / ln_gsdp	0.0362	0.0618	0.562	350
state	base	mu_med / lhhi	-0.0096	0.1518	0.950	350
state	growth	mu_med / gsdp-gr	-0.0000	0.0004	0.960	317
state	percap	mu_med / ln_pcnsdp	0.0645	0.0580	0.275	350
state	noCOVID	mu_med / ln_gsdp	0.0261	0.0615	0.674	284
state	lag	mu_med / L.ln_gsdp	0.0484	0.0636	0.452	321

Panel	Spec	Outcome / regressor	Coef.	s.e.	<i>p</i>	<i>N</i>
state	base	psi_med / ln_gsdp	-0.0513	0.1238	0.681	350
state	base	psi_med / lhhi	-0.4438	0.2478	0.083	350
state	growth	psi_med / gsdp_gr	0.0016	0.0010	0.127	317
state	percap	psi_med / ln_pcnsdp	-0.0414	0.1006	0.684	350
state	noCOVID	psi_med / ln_gsdp	-0.0478	0.1318	0.719	284
state	lag	psi_med / L.ln_gsdp	-0.1949	0.1124	0.093	321
state	base	monoshare / ln_gsdp	-0.0824	0.0681	0.236	350
state	base	monoshare / lhhi	0.0674	0.1042	0.523	350
state	growth	monoshare / gsdp_gr	0.0004	0.0006	0.510	317
state	percap	monoshare / ln_pcnsdp	-0.0737	0.0616	0.240	350
state	noCOVID	monoshare / ln_gsdp	-0.0949	0.0655	0.157	284
state	lag	monoshare / L.ln_gsdp	-0.1024	0.0721	0.165	321
state	base	vartfpr100 / ln_gsdp	39.0816	17.1790	0.030	350
state	base	vartfpr100 / lhhi	33.2314	45.5319	0.471	350
state	growth	vartfpr100 / gsdp_gr	-0.0787	0.2736	0.775	317
state	percap	vartfpr100 / ln_pcnsdp	37.7393	19.7495	0.065	350
state	noCOVID	vartfpr100 / ln_gsdp	44.3816	16.0768	0.009	284
state	lag	vartfpr100 / L.ln_gsdp	30.0716	18.3203	0.111	321